

Question 1

(a)

To show T is a basis for V , we verify two properties:

1. Linear Independence: Form the matrix with vectors in T as columns:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 3 & -1 \end{pmatrix}$$

The first three rows form the identity matrix, which has rank 3. Since the columns are linearly independent in $\mathbb{R}^4$, T is linearly independent.

2. Spanning: Each vector in T must belong to V . For the first vector $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix}$, solve:

$$\begin{cases} 2p + 2q + r = 1 \\ p + q - r + 3s = 0 \\ p + q + 2r = 0 \\ 6p + 6q - 3r + 9s = 2 \end{cases}$$

Solving gives $p = \frac{2}{3}, q = -\frac{1}{3}, r = -\frac{1}{3}, s = -\frac{1}{3}$, confirming it belongs to V . Similar checks for the other vectors confirm they are in V . Since $\dim(V) = 3$ and T has 3 linearly independent vectors, T spans V . Thus, T is a basis.

(b)

Given coordinates $[v]_T = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$, the vector v is:

$$v = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 3 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ 2x_1 + 3x_2 - x_3 \end{pmatrix}$$

(c)

Gram-Schmidt Process:

• **Step 1:** Compute $\mathbf{u}_1$:

$$\mathbf{u}_1 = \mathbf{t}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix}, \quad \|\mathbf{u}_1\| = \sqrt{1^2 + 2^2} = \sqrt{5}, \quad \mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix}$$

- **Step 2:** Compute $\mathbf{u}_2$:

$$\text{proj}_{\mathbf{u}_1}(\mathbf{t}_2) = \frac{\mathbf{t}_2 \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 = \frac{3}{\sqrt{5}} \mathbf{e}_1 = \begin{pmatrix} \frac{3}{5} \\ 0 \\ 0 \\ \frac{6}{5} \end{pmatrix}$$

$$\mathbf{u}_2 = \mathbf{t}_2 - \text{proj}_{\mathbf{u}_1}(\mathbf{t}_2) = \begin{pmatrix} -\frac{3}{5} \\ 1 \\ 0 \\ \frac{9}{5} \end{pmatrix}, \quad \|\mathbf{u}_2\| = \sqrt{\frac{9}{25} + 1 + \frac{81}{25}} = \sqrt{\frac{115}{25}} = \sqrt{\frac{23}{5}}$$

$$\mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{23/5}} \begin{pmatrix} -\frac{3}{5} \\ 1 \\ 0 \\ \frac{9}{5} \end{pmatrix}$$

- **Step 3:** Compute $\mathbf{u}_3$:

$$\text{proj}_{\mathbf{u}_1}(\mathbf{t}_3) = \frac{\mathbf{t}_3 \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 = \frac{-2}{5} \mathbf{u}_1, \quad \text{proj}_{\mathbf{u}_2}(\mathbf{t}_3) = \frac{\mathbf{t}_3 \cdot \mathbf{u}_2}{\|\mathbf{u}_2\|^2} \mathbf{u}_2$$

After calculations:

$$\mathbf{u}_3 = \mathbf{t}_3 - \text{proj}_{\mathbf{u}_1}(\mathbf{t}_3) - \text{proj}_{\mathbf{u}_2}(\mathbf{t}_3) = \begin{pmatrix} \frac{1}{7} \\ \frac{3}{14} \\ 1 \\ -\frac{1}{14} \end{pmatrix}, \quad \|\mathbf{u}_3\| = \sqrt{\frac{15}{14}}, \quad \mathbf{e}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|}$$

Orthonormal basis $S = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$.

(d)

Coordinates $[v]_S$ are the coefficients when v is expressed in terms of S . Using the orthonormal basis:

$$[v]_S = \begin{pmatrix} \mathbf{e}_1 \cdot v \\ \mathbf{e}_2 \cdot v \\ \mathbf{e}_3 \cdot v \end{pmatrix} = \begin{pmatrix} \sqrt{5}x_1 + \frac{6}{\sqrt{5}}x_2 - \frac{2}{\sqrt{5}}x_3 \\ \sqrt{\frac{14}{5}}x_2 - \frac{3}{\sqrt{70}}x_3 \\ \sqrt{\frac{15}{14}}x_3 \end{pmatrix}$$

(e)

Project $\mathbf{w}$ onto V . Using the basis T :

$$[w_p]_T = (\mathbf{w} \cdot \mathbf{t}_1, \mathbf{w} \cdot \mathbf{t}_2, \mathbf{w} \cdot \mathbf{t}_3) = \begin{pmatrix} -1(1) + 5(2) \\ -2(1) + 5(3) \\ 2(1) + 5(-1) \end{pmatrix} = \begin{pmatrix} 9 \\ 13 \\ -3 \end{pmatrix}$$

Normalizing gives $[w_p]_T = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Question 2

Month	Advertising Spend (X)	Sales (Y)
1	500	4
2	700	6
3	800	7
4	600	5.5
5	1000	8

Compute Required Summations

$$\sum X_i = 500 + 700 + 800 + 600 + 1000 = 3600,$$

$$\sum Y_i = 4 + 6 + 7 + 5.5 + 8 = 30.5,$$

$$\sum X_i Y_i = (500 \times 4) + (700 \times 6) + (800 \times 7) + (600 \times 5.5) + (1000 \times 8) = 23,100,$$

$$\sum X_i^2 = 500^2 + 700^2 + 800^2 + 600^2 + 1000^2 = 2,740,000.$$

Calculate Slope m and Intercept b

$$m = \frac{n \sum X_i Y_i - \sum X_i \sum Y_i}{n \sum X_i^2 - (\sum X_i)^2} = \frac{5 \times 23,100 - 3600 \times 30.5}{5 \times 2,740,000 - (3600)^2} = \frac{5,700}{740,000} \approx 0.0077,$$

$$b = \frac{\sum Y_i - m \sum X_i}{n} = \frac{30.5 - 0.0077 \times 3600}{5} \approx \frac{2.78}{5} \approx 0.556.$$

Solve for X When $Y = 5$

$$X = \frac{5 - 0.556}{0.0077} \approx \frac{4.444}{0.0077} \approx 577.15 \Rightarrow \boxed{577}.$$

Final Answer: $\boxed{577}$ dollars.

Question 3

(a)

Define $u_1 = y_1$, $u_2 = y_1'$, $v_1 = y_2$, $v_2 = y_2'$. The system becomes:

$$\begin{cases} u_1' = u_2 \\ u_2' = -2u_2 + 2u_1 \\ v_1' = v_2 \\ v_2' = -4v_2 - 5v_1 \end{cases}$$

Matrix form:

$$\mathbf{z}' = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -5 & -4 \end{pmatrix} \mathbf{z}, \quad \mathbf{z} = \begin{pmatrix} u_1 \\ u_2 \\ v_1 \\ v_2 \end{pmatrix}$$

(b)

For y_1 : Characteristic equation: $r^2 + 2r - 2 = 0$ Roots: $r = -1 \pm \sqrt{3}$ General solution:
 $y_1(t) = C_1 e^{(-1+\sqrt{3})t} + C_2 e^{(-1-\sqrt{3})t}$

For y_2 : Characteristic equation: $r^2 + 4r + 5 = 0$ Roots: $r = -2 \pm i$ General solution:
 $y_2(t) = e^{-2t}(D_1 \cos t + D_2 \sin t)$

(c)

Apply initial conditions:

$$y_1(0) = 0 \Rightarrow C_1 + C_2 = 0, \quad y_1'(0) = \sqrt{3} \Rightarrow (-1 + \sqrt{3})C_1 + (-1 - \sqrt{3})C_2 = \sqrt{3}$$

Solving gives $C_1 = \frac{1}{2}$, $C_2 = -\frac{1}{2}$.

For y_2 :

$$y_2(0) = -1 \Rightarrow D_1 = -1, \quad y_2'(0) = 1 \Rightarrow -2D_1 + D_2 = 1 \Rightarrow D_2 = -1$$

Particular solution: $y_1(t) = \frac{1}{2}e^{(-1+\sqrt{3})t} - \frac{1}{2}e^{(-1-\sqrt{3})t}$ $y_2(t) = -e^{-2t}(\cos t + \sin t)$